
JOSÉ DANIEL VELÁSQUEZ CARMONA

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EDUCATION

Doctor of Science in Chemistry - Physical Chemistry August 2016 – December 2021
Universidad del Valle, Cali-Colombia

Research project: *'Synthesis, functionalization and characterization of carbon nanooxions for the development of materials with application in supercapacitance'*, SIMERQO Research Group, Advisor: Manuel Noe Chaur.

Bachelor of Science in Chemistry February 2010 – June 2015
Universidad del Valle, Cali-Colombia

Dissertation title: *'Solar disinfection by three reactors using photoFenton process at neutral pH'*, GAOX Research Group, Advisor: Luis Norberto Benítez.

Chemistry Technologist August 2007 – June 2010
Universidad del Valle, Cali-Colombia

RESEARCH EXPERIENCE

- Research in fullerene derivative materials for supramolecular and photovoltaic applications. Carbon materials laboratory – University of Texas at El Paso. Dr Luis Echegoyen
- Research in Synthesis, functionalization, and characterization of carbon allotropes for energy storage devices and catalysis.
- Research in PhotoFenton process for water disinfection at neutral pH.

ACADEMIC EXPERIENCE

- Adjunct Professor: Theoretical and practical general and Physical chemistry for bachelor students.
Supervisor: Dr.Sc. Norberto Benítez
Universidad del Valle
August 2021 – Currently
- Adjunct Professor: Theoretical and practical Physical chemistry and Biochemistry for bachelor students.
Supervisor: Dr.Sc. Jimmy Morales
Universidad Santiago de Cali
February 2022 – Currently

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- High School Teacher: Science and Chemistry

Supervisor: María Alejandra Quintero
New Cambridge School – Bilingual School
August 2021 – Currently

- Assistant Professor: Theoretical and practical Physical and Organic chemistry for bachelor students.

Supervisor: Dr. Sc. Luis Norberto Benítez
Universidad del Valle, Cali-Colombia
August 2016 – December 2020

- Technical Institute Professor: Chemistry, Physics and Math

Supervisor: Catalina Campo
Academia de Aviación y Turismo Internacional
August 2014 – August 2015

Poster and oral presentations

- Functionalization of carbon nanooxions with pyrene moieties for possible applications in supercapacitors and transporting layers. 4th International Congress Next Generation Solar Energy, held in Cali-Colombia. 2017
- Synthesis, functionalization, and Characterization of carbon nanooxions for the development of supercapacitors. VI Nanoandes Nanotechnology School, held in Cali-Colombia. 2016
- Disinfection in water by the enhanced SODIS method at neutral pH. IV Chemistry Symposia, Department of Chemistry, Universidad del Valle, Cali-Colombia. 2014

SKILLS

- Good ability to design synthesis routes for organic molecules.
- Good technical skills to conduct air and moisture sensitive reactions.
- Experience in the synthesis of carbon allotropes and its functionalization.
- Practical and theoretical knowledge of common techniques used in the characterization of organic molecules and surfaces (Raman, TGA, SEM, IR, MALDI-TOF, XRD, NMR and UV-Vis).
- Good ability in electrochemical techniques (Cyclic Voltammetry and Electrochemical Impedance Spectroscopy).
- Abilities using Fortran 77 and Gaussian 09W for molecular simulation.
- Pedagogical aptitudes for developing the acquisition of knowledge.

LANGUAGE SKILLS

Spanish: Native, English: B2

HONORS AND AWARDS

- Dean's list student in bachelor's degree
- Award for teaching with TICs (information and communication technologies)

PUBLICATIONS

- Grande Tovar, C.D., Castro, J.I., Valencia, C.H., Navia Porras, D.P., Mina Hernandez, J.H., Valencia, M.E., Velásquez, J.D. and Chaur, M.N., 2019. Preparation of chitosan/poly (vinyl alcohol) nanocomposite films incorporated with oxidized carbon nano-onions (multi-layer fullerenes) for tissue-engineering applications. *Biomolecules*, 9(11), p.684.

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- Velásquez, J.D., Tomczykowa, M., Plonska-Brzezinska, M.E. and Chaur, M.N., 2020. Evaluation of the covalent functionalization of carbon nano-onions with pyrene moieties for supercapacitor applications. *Materials*, 13(5), p.1141.
 - Velasquez, J.D., Rubio, J.C. and Chaur, M.N., 2020. Functionalization of multilayer fullerenic nanostructures with l-lysine for supercapacitor applications: Comparing oxidation against diazonium salt methods. *Diamond and Related Materials*, 105, p.107771.
 - Castro, E., Fernandez-Delgado, O., Artigas, A., Zavala, G., Liu, F., Moreno-Vicente, A., Rodríguez-Forte, A., Velasquez, J.D., Poblet, J.M. and Echegoyen, L., 2020. α -DTC 70 fullerene performs significantly better than β -DTC70 as electron transporting material in perovskite solar cells. *Journal of Materials Chemistry C*, 8(20), pp.6813-6819.
 - Ortiz-Restrepo, J.E., Loaiza, O.A., Urresta, J.D., Velasquez, J.D., Pastor, E., Chaur, M.N. and Lizcano-Valbuena, W.H., 2021. A comparative study of different carbon materials as metal-free catalysts for oxygen reduction and hydrogen evolution reactions in alkaline media. *Diamond and Related Materials*, 117, p.108464.
 - Duarte-Ruiz, A., Torres-Cortés, S.A., Meléndez, A., Velásquez, J.D. and Chaur, M.N., 2021. Synthesis and characterization of C60 and C70 acetylacetone monoadducts and study of their photochemical properties for potential application in solar cells. *Revista Colombiana de Química*, 50(1), pp.86-97.
 - In progress: Synthesis of concentric fullerenic nanostructures from polycrystalline coal. Velasquez, J. D., Cabal, K., Barraza, J.M., Plonska-Brzezinska, M.E, Rubio, J.C., Chaur, M.N.